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Electronic Nose and GC-MS Analysis to Detect Mango Twig Tip Dieback in Mango (*Mangifera indica*) and Panama Disease (TR4) in Banana (*Musa acuminata*)

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Abstract: Volatile organic compounds (VOCs), as a biological element released from plants, have been correlated with disease status. Although analysis of VOCs using GC-MS is a routine procedure, it has limitations, including being time-consuming, laboratory-based, and requiring specialist training. Electronic nose devices (E-nose) provide a portable and rapid alternative. This is the first pilot study exploring three types of commercially available E-nose to assess how accurately they could detect mango twig tip dieback and Panama disease in bananas. The devices were initially trained and validated on known volatiles, then pure cultures of *Pantoea* sp., *Staphylococcus* sp., and *Fusarium odoratissimum*, and finally, on infected and healthy mango leaves and field-collected, infected banana pseudo-stems. The experiments were repeated three times with six replicates for each host-pathogen pair. The variation between healthy and infected host materials was evaluated using inbuilt data analysis methods, mainly by principal component analysis (PCA) and cross-validation. GC-MS analysis was conducted contemporaneously and identified an 80% similarity between healthy and infected plant material. The portable C 320 was 100% successful in discriminating known volatiles but had a low capability in differentiating healthy and infected plant substrates. The advanced devices (PEN 3/MSEM 160) successfully detected healthy and diseased samples with a high variance. The results suggest that E-noses are more sensitive and accurate in detecting changes of VOCs between healthy and infected plants compared to headspace GC-MS. The study was conducted in controlled laboratory conditions, as E-noses are highly sensitive to surrounding volatiles.

Keywords: electronic nose; headspace gas chromatography-mass spectrometry (GC-MS) analysis; principal component analysis (PCA); volatile organic compounds (VOC); Panama disease; tropical race 4 (TR4); mango twig tip dieback (MTTD)



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1. Introduction

Detecting plant infections before visible symptoms appear is crucial for implementing effective management and pest control measures to curb disease spread [1,2]. However, it is often challenging to identify physical signs of infection in plants during the asymptomatic or early stages of infestation [3]. The conventional approaches for the diagnosis of plant diseases are laboratory-based methods, such as polymerase chain reaction (PCR), immunofluorescence (IF), fluorescence in situ hybridisation (FISH), enzyme-linked immunosorbent assay (ELISA) and flow cytometry (FCM) [4,5]. These techniques are expensive, time-consuming, require specialist training, and have limited application for monitoring large farms where several thousand plants are cultivated. To achieve mass monitoring, non-invasive technologies should be used [6]. Recent technology focusing on detecting VOCs released from plants has gained attention for its non-invasiveness and rapid results [7].

Plants emit many VOCs, which deliver functional information related to their growth, health, and disease [8]. As a defence mechanism, plants emit repellent VOCs to deter insect attacks. VOCs play significant roles in plant communication, promising enhanced crop protection [9,10]. Their composition varies based on damage type, such as pathogen infection or herbivore feeding [11]. VOC profiles, termed “VOC fingerprints”, indicate real-time physiological status and offer bio-information for early-stage diagnosis [12].

Generally, VOCs are detected and analysed using a combination of a collection method (e.g., Solid Phase Micro-Extraction—SPME) alongside an analytical chemistry method (e.g., GC-MS) [3]. SPME is a solvent-free sampling technique allowing non-exhaustive volatile extraction [13] and using a small analyte quantity compared to traditional solvent-based methods [14]. Analytes collected from SPME are analysed by GC-MS for the identities and quantities [15], displaying remarkable potential in plant diagnostic applications. Despite the accuracy of quantitative and qualitative analysis, GC-MS analysis is not always readily available and affordable [16].

However, GC-MS is time-consuming and requires laboratory-based testing, which limits its real-time in-field application. For effective disease management using plant VOC data, portable, user-friendly, real-time detection technology is needed. E-noses, or artificial intelligent noses (AI noses), are designed to mimic the functionality of a natural olfactory system. E-noses can provide a rapid and real-time approach to detecting VOCs [17]. They utilise different sensor arrays, signal conditioning modules, data acquisition units, and pattern recognition algorithms [18].

The E-nose does not provide quantification or identification of specific VOCs like GC-MS. Studies such as the characterisation of odour-active compounds in Californian chardonnay wines using GC-olfactometry and GC-mass spectrometry [19] and GC-MS analysis of breath odour compounds in liver patients [20] reveal the ability of GC-MS to identify specific VOCs that E-noses are not capable of. However, it is an easily operated, fast-turnaround tool for recognising differences among VOC samples (e.g., healthy vs infected), such as detecting and identifying fungal species [21]. Also, the same device can be used for many different applications, e.g., food freshness, quality, ripeness, and shelf-life [22].

E-noses have been used as a non-invasive technique and have been applied for food safety and quality analysis [23]. More recently, the devices have been used to detect fungi and oomycetes [24], and the early detection of *Fusarium wilt* of banana “Panama disease” [6] caused by the soil-borne fungus *Fusarium odoratissimum* (syn. *F. oxysporum* f.sp. *cubense*). The strain “Tropical Race 4” was first discovered in Darwin, Northern Territory (NT), in 1999 and Tully, Queensland, Australia, in 2015 [25].

Mango Twig Tip Dieback (MTTD) describes an apical necrosis of mango shoots. Severity varies from orchard to orchard across ages, with Kensington Pride (KP) and Nam Doc Mai cultivars being the most susceptible [Mango Twig Tip Dieback | Department of Industry, Tourism and Trade, <https://industry.nt.gov.au/publications/primary-industry-publications/newsletters/regional-newsletters/rural-review/nt-rural-review-november-2021/mango-twig-tip-dieback>, accessed on 5 January 2024]. Two bacterial pathogens, *Pantoea* sp. and *Staphylococcus* sp., have been associated with the disorder, but the exact aetiology of MTTD remains unknown.

2. Materials and Methods

A series of experiments were conducted to explore the capability of three E-nose devices to identify TR4 in banana pseudo-stems and MTTD in mango leaves. Inoculated substrates, samples from field collections, and healthy specimens with no observable symptoms were trialled with both E-noses and GC-MS analysis to confirm VOC fingerprints.

2.1. Electronic Nose Devices

The Cyranose 320 (C 320-Sensigent LLC, Baldwin Park, CA, USA) features the NoseChip™ with 32 polymer composite chemi-resistor sensors coated with conductive

films arranged across electrodes. The Portable Electronic Nose 3 (PEN 3) by Win Muster Airsense (WMA) Analytics Inc. (Schwerin, Germany) combines a sampling apparatus, a detector unit with an array of sensors, and pattern recognition software (Win Muster v.1.6) for data recording and analysis. The MSEM 160 (Sensigent LLC, Baldwin Park, CA, USA) is a multi-sensor and portable instrument for measuring chemicals and odours. This instrument consists of nanocomposite sensors, metal oxide semiconductors (MOS), and electrochemical and photoionisation sensors. The MSEM 160 is controlled by an internal computer running the Windows or Linux operating system.

2.2. SPME Coupled with GC-MS

The experiments also utilised sampling with SPME portable Field Sampler, coating CAR/PDMS (Sigma, Bayswater, VIC, Australia) and GC/MS analysis: an Agilent 88890 model with an Agilent 59777B single quadrupole electron ionisation mass selective detector using Agilent MassHunter Workstation GC/MS Data Acquisition (version 10.1.49) software. GC/MS data were analysed using Agilent MassHunter Workstation Qualitative Analysis (version 10.0) software. Chemical identification of VOC peaks detected was performed by searching the NIST/EPA/NIH Mass Spectral Library (Version 3.0). Identification was regarded as preliminary for spectra matched in the NIST library with a similarity above 70%. The preliminary identification was further confirmed by searching the literature that reported these compounds from similar sources (i.e., VOCs identified from mango and banana).

2.3. Assessing E-Nose Functions and Sampling on Known Volatiles

Isoamyl isovalerate (IAIV) and 3-methyl-2-butanol, 98% (3M2B) were obtained from Thermo Fisher Scientific Pty Ltd., Scoresby, VIC, Australia, and a dilution series of five concentrations (0, 1, 10, 100, 1000 ppm) prepared, using n-hexane, 95% as the solvent and the negative control. Mixtures of the two compounds (1 ppm and 100 ppm) were also prepared. Two ml of each was transferred into a 20 mL headspace glass vial and was allowed to saturate for six hrs before being assessed using the three E-nose devices. Five replicates were prepared for each mixture.

The experimental design shown in Figure 1 outlines the method applied in the differentiation of plant and pathogen VOC signatures using E-noses.

2.4. Pure Cultures of Plant Pathogens and Controls for Training

F. odoratissimum (TR4) was isolated from infected cavendish pseudo-stems and pure cultures were produced on potato dextrose agar (PDA). Pure cultures of *F. odoratissimum* were then subcultured to PDA, hyphal tip cultures were obtained, and identity was confirmed by PCR. Petri dishes were sealed with parafilm and incubated at 18 °C for seven days. Uninoculated PDA media in Petri dishes were prepared as the control. Two pathogens of MTTD 18—*Pantoea* sp. and MTTD 30—*Staphylococcus* sp. were recovered from mango twigs from Colton Road, Northern Territory, Australia. MTTD isolates were obtained on nutrient agar (NA) that were sub-cultured to obtain single colony cultures, and identity was confirmed by PCR. Petri dishes were sealed with parafilm and incubated at 18 °C for seven days. Uninoculated NA media in Petri dishes were prepared as the control. TR4 cultures grown on PDA, *Pantoea* sp. and *Staphylococcus* sp. cultures grown on NA in Petri dishes. Petri dishes were half-opened and secured with biofilm for rapid collection of VOCs into gas-collecting bags. VOCs were collected by inserting the SPME fibres into the bags in duplicates for 17 h. VOCs from uninoculated PDA and NA, clean Petri dishes and empty gas-collecting bags were also collected to quantify any background VOCs. Further SPME fibers were subjected to GC-MS analysis.

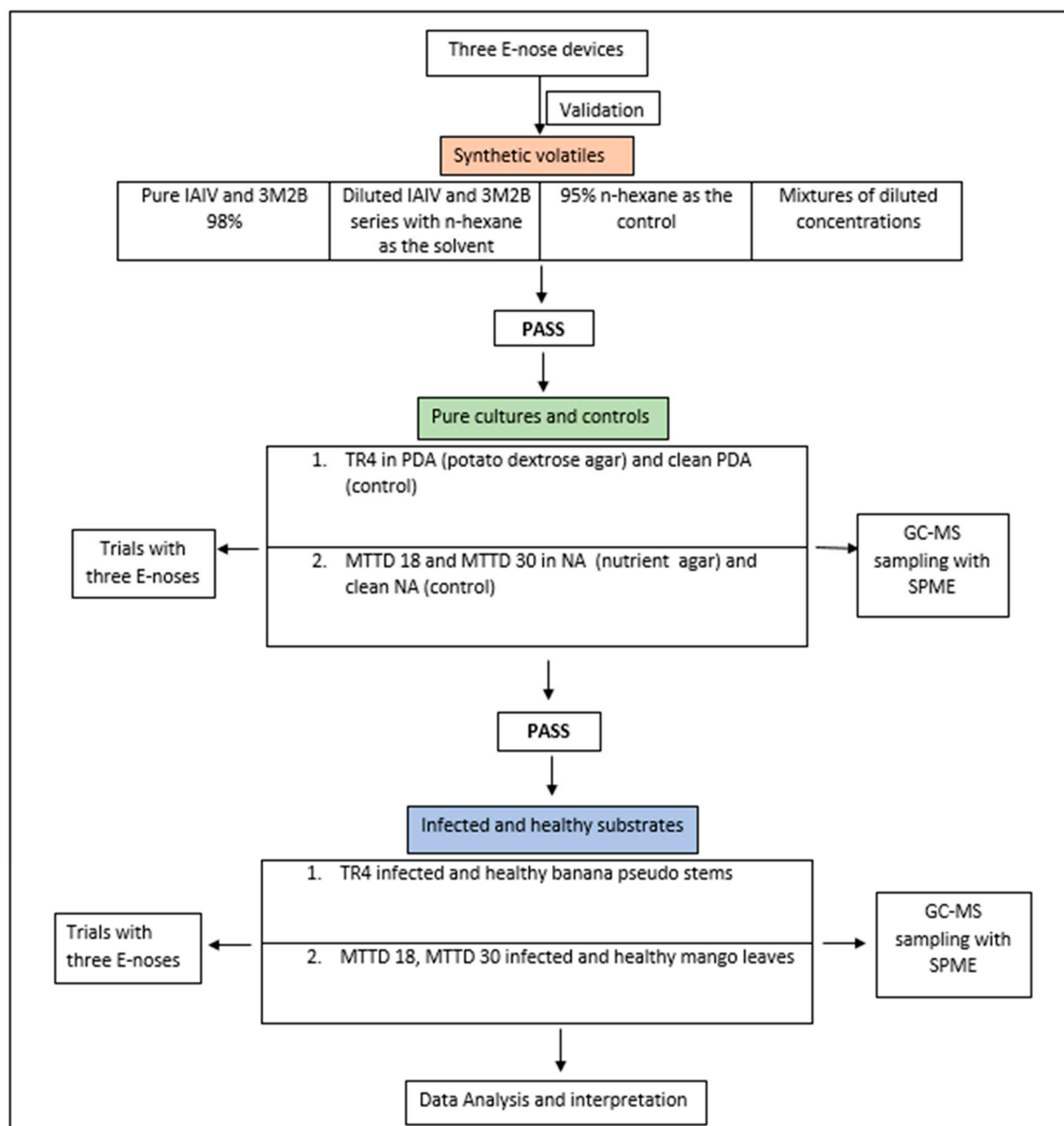


Figure 1. An overview of the experimental design.

TR4 cultures on PDA were cut into thin strips, then placed in six septum sealable, 20 mL headspace glass vials and incubated for 24 h at room temperature. Headspace sampling of vials was conducted with three E-nose devices after 24 h of saturation. Procedures were repeated with two MTTD bacterial cultures, uninoculated NA and PDA media.

2.5. Sampling of Healthy and Infected Substrates

To infect mango leaves with MTTD pathogens, healthy ones were washed with running tap water and dried. The leaves were trimmed from the edge of the petiole to fit into a petri dish. Inoculation steps were carried out inside the laminar flow cabinet. The surface of each leaf was sterilised with 70% ethanol for one minute and then washed in distilled water for one minute. The leaves were dried with sterilised filter papers and kept inside clean Petri dishes. A surface sterilised, 12 mm cork borer was used to extract MTTD 18 and 30 from the petri dish and pushed into a mango leaf to make a wound in the epidermal

tissue. Two punches were made on each leaf, and the leaves were placed in a clean Petri dish and wrapped with biofilm. Fifteen inoculated leaves in individual petri dishes were prepared separately for MTTD 18 and MTTD 30 and incubated at 25 °C for five days.

After five days of incubation, each petri dish was half-opened and enclosed in gas-collecting bags. Four petri dishes were included in each gas-collecting bag, and two gas-collecting bags were prepared. Bags were placed in the incubator for another 24 h at 25 °C. Sterilised distilled water was used as the control.

After 24 h of incubation, SPMEs were inserted into gas collecting bags for 17 h from the control and infected mango leaves with MTTD 18 and MTTD 30 and subjected to GC-MS analysis. Six mango leaves in the control set were sliced and placed individually in six 20 mL headspace vials with sealed septa. Before E-nose sampling, these vials were allowed to rest for 24 h at room temperature to saturate headspace. Procedures were repeated with infected mango leaves with two MTTD bacterial cultures. E-nose sampling was carried out for eighteen replicates of control and infected samples (Figure 2).

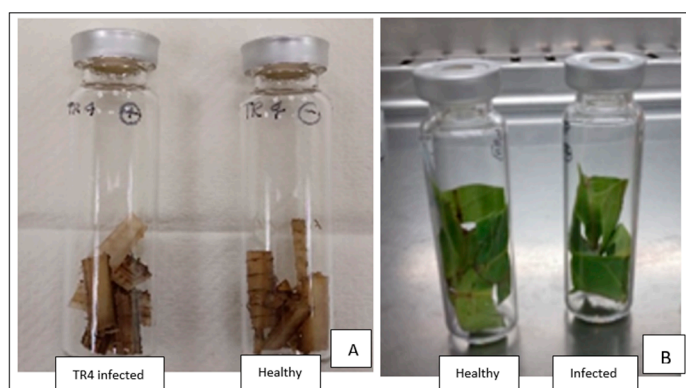


Figure 2. Thin strips of substrates were placed in 20 mL headspace glass vials prior to E-nose sampling. (A) Healthy and TR4-infected banana pseudo stems and (B) healthy and infected mango leaf samples.

Thin strips of healthy banana pseudo-stems (received from the field) were placed in six replicates of 20 mL headspace glass vials. TR4 positive (according to the external appearance of purple lines along the vascular tissues and confirmed by sterile culturing) pseudo-stems followed the same procedure. All replicates were allowed to incubate for 24 h to saturate the headspace, and E-nose sampling was carried out on the three devices. Two pieces (10 cm × 3 cm each) of healthy banana pseudo-stems were placed in collecting gas-collecting bags following a heat seal of the opening in duplicates. The procedure was repeated for TR4-infected pseudo-stems. VOCS were collected using SPMEs, as explained in the earlier section and subjected to GC-MS analysis.

2.6. Data Analysis

Inbuilt statistical analysis methods were utilised for three E-noses: Win Muster v.1.6 in PEN 3, CDA software, v.2.39 in MSEM 160, and inbuilt PC nose software in C 320. These tools were employed for data analysis via principal component analysis (PCA), a common pattern recognition method in E-nose analysis. PCA transforms the original dataset of correlated variables into linearly uncorrelated principal components [26], effectively reducing dimensions while retaining valuable information [27].

CDA software in MSEM 160 instrument monitors changes in the electrical, physical, optical and/or chemical properties of the internal arrays of chemical sensors. Changes in the sensor properties were identified and qualified by the software and translated into metrics during operation according to the explanation in the MSEM 160 user manual.

PC nose software was employed for analysing C 320 data through cross-validation, which is a statistical technique used to evaluate and compare learning algorithms by dividing data into two segments: one for training the model and the other for validation.

In typical cross-validation, the training and validation sets overlap in successive rounds, allowing each data point to be validated against others [28].

3. Results and Discussion

3.1. Discrimination Capability of Known Volatiles

The C 320 device was unsuccessful in differentiating between the concentration gradient of 3M2B and IAIV (Table 1), demonstrating how many replicates were identified successfully out of five samples for each subsequent concentration of both 3M2B and IAIV. Limitations in sensitivity resulted in poor discrimination in each subsequent concentration of the same compound.

Table 1. Cross-validation results for each of the five replicates of different concentrations of 3M2B and IAIV in C 320.

Trained as		0 ppm	1 ppm	10 ppm	100 ppm	1000 ppm	% Accuracy
No. of correct identification/5 (total no. of replicates)	3M2B	3	1	1	4	4	52
	IAIV	0	4	0	3	3	40

A successful differentiation was observed between pure solutions of water, 95% n-hexane and 70% ethanol (Table 2). These results suggest that Cyranose 320 is more suitable for separating completely different samples than compounds with similar volatile characteristics.

Table 2. Cross-validation results for pure water, 95% n-Hexane and 70% ethanol in C 320.

Trained as	Water	95% n-Hexane	70% Ethanol	% Accuracy
No. of correct identification/5 (total no. of replicates)	5	5	5	100

The PEN 3 could differentiate all five classes of concentrations (0, 1, 10, 100, 1000 ppm) of both 3M2B and IAIV (Supplementary Material Figure S1A). Five separated clusters were clearly observed in the PCA plots, with a variance of 87.59% for 3M2B and 88.05% for IAIV. A standard pattern was created from Winmuster software by merging data for all five subsequent concentrations. This was called the “Total Concentration Series”. The two extreme concentrations (0 and 1000 ppm) were located apart from the total series with low percentages of variance. This was observed in both the 3M2B and IAIV series. This result identifies PEN 3 as more efficient than C 320 in differentiating consecutive concentrations of the same compound.

PEN 3 was then trialled to differentiate between two compounds, 3M2B and IAIV. A distinct separation of the two classes was observed with a high variance of 87.16% (Supplementary Material Figure S2A). PEN 3 could discriminate between not only the different compounds but also consecutive concentrations of the same compound. Mixtures of the compounds with optimum concentrations (1 ppm and 100 ppm) were successfully differentiated from pure compounds with a high variance of 86.43% (Supplementary Material Figure S2B).

The MSEM 160 was successful in discriminating consecutive concentrations of compounds (Supplementary Material Figure S3A). CD analysis was carried out to convert raw data into a score plot in PCA. Supplementary Figure S3B shows the ability of MSEM 160 to differentiate two compounds in the mixture of 1 ppm 3M2B and IAIV.

3.2. Efficacy of Discriminating Pure Cultures of Plant Pathogens and Controls

3.2.1. Electronic Noses

Twelve replicates of TR4 cultures were divided into two groups, “TR4–G 1” and “TR4–G 2”, with six replicates in each. C 320 successfully identified all 12 samples as TR4,

although confusion between “G 1” and “G 2” occurred (Table 3). None of the replicates were failed to detect as TR4 and not presented as unknown. Results express that C 320 could differentiate the VOCs of *F. odoratissimum* cultures from PDA. C 320 was 77.78% successful in detecting eighteen replicates of MTTD cultures and controls. Table 4 demonstrates that all six controls were identified as NA-control while confusion occurred between two MTTD bacterial varieties. Similar volatile characteristics of *Pantoea* sp. and *Staphylococcus* sp. might affect the indistinct separation of the two sample groups. The results indicate that C 320 is more convenient for detecting compounds with entirely different volatile profiles.

Table 3. C 320 cross-validation results for sampling two sets of TR4 cultures.

Trained as	TR4-G 1	TR4-G 2	% Accuracy
No. of correct identification/6 (total no. of replicates)	5	5	83.33

Table 4. C 320 cross-validation results for sampling uninoculated NA, MTTD 18 and MTTD 30 cultures.

Trained as	NA-Control	MTTD-18	MTTD-30	% Accuracy
No. of correct identification/6 (total no. of replicates)	6	4	4	77.78

PCA projection plot in Figure 3 demonstrates that MTTD 18 and MTTD 30 have similar volatile characteristics, which differ from the uninoculated NA control. Therefore, samples of two MTTD varieties were clustered on the same plane opposite to the control samples.

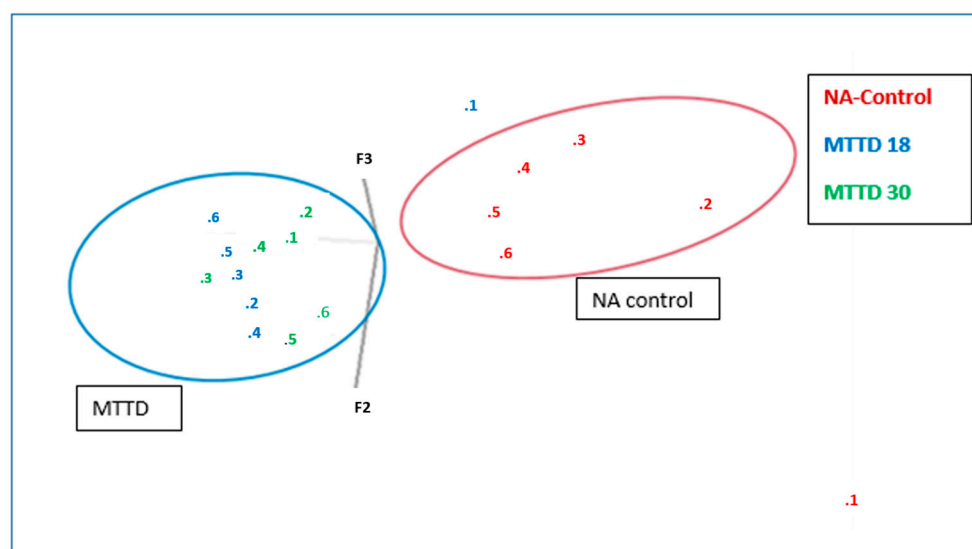


Figure 3. PCA plot for two MTTD bacterial varieties and NA control in C 320.

PCA plot for TR4 culture and uninoculated PDA controls using PEN 3 exhibited a 72.45% variance in the first main axis but a low variance of 15.52% in the second main axis, as shown in Figure 4A. This demonstrates that PEN 3 was less capable of differentiating volatile organic compounds of TR4 cultures from uninoculated PDA. The results demonstrate that PDA volatiles have some impact on TR4 volatile characteristics, hence giving two overlapping clusters. Distinct separation was observed between three classes (NA control, MTTD 18 and MTTD 30) with a high variance of 92.26%, as shown in Figure 4B. Samples of the same class have been clustered together, showing similar patterns after analysing raw data through “Winmuster” software, v.1.6. Different volatile compounds between each class would lead to different patterns and, hence, distinct separation of the classes.

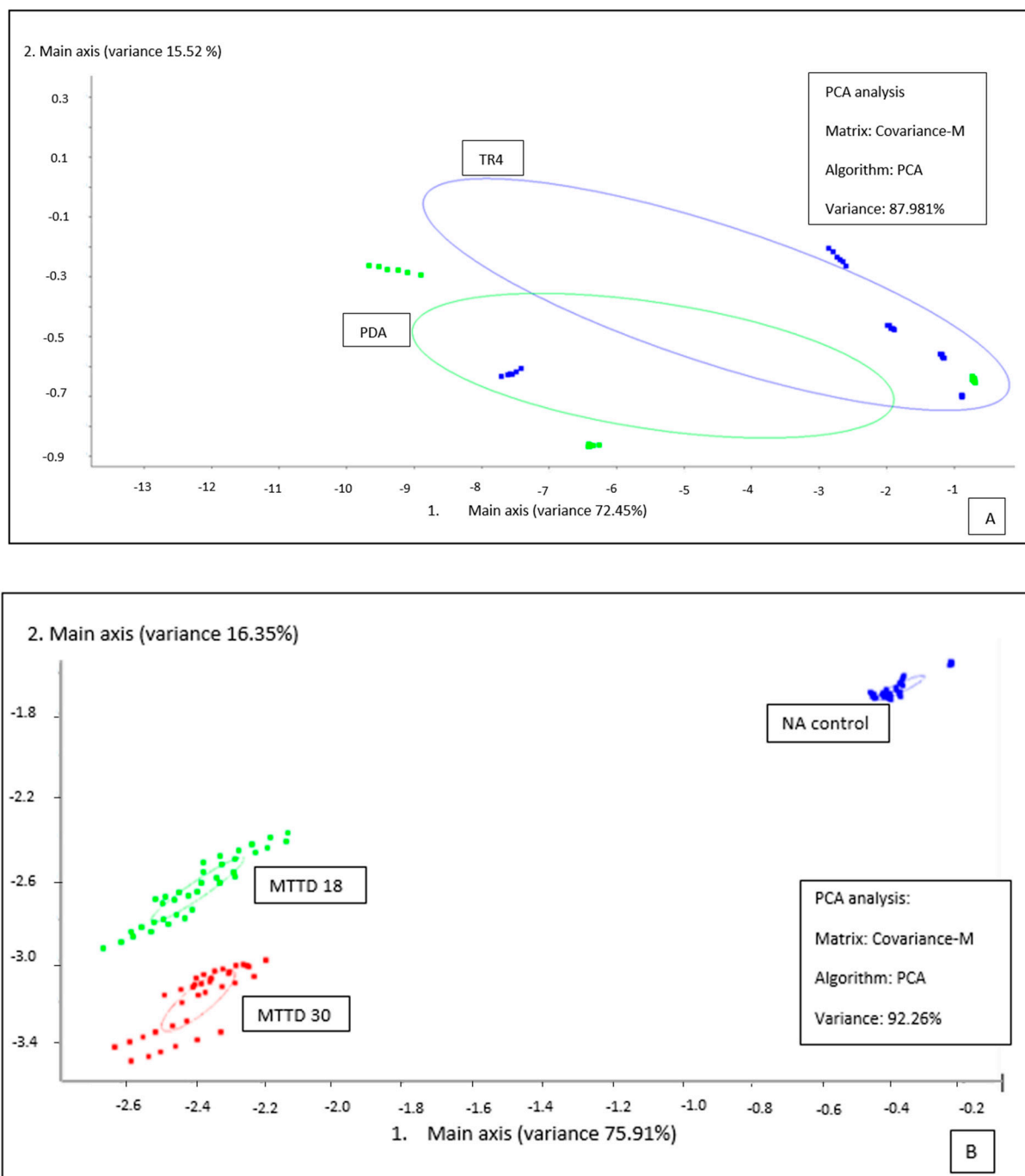


Figure 4. (A) PCA plot for TR4 culture and PDA control, (B) PCA plot for MTTD 18, MTTD 30 pure cultures and uninoculated NA control in PEN 3.

Figure 5A demonstrates that the MSEM 160 distinct separation between TR4 cultures and PDA controls. Two classes were separated and clustered independently, which provides statistical evidence for the high efficacy of MSEM 160. Figure 5B clearly shows the efficiency of the MSEM 160 in discriminating between MTTD 18 and MTTD 30 cultures and NA control.

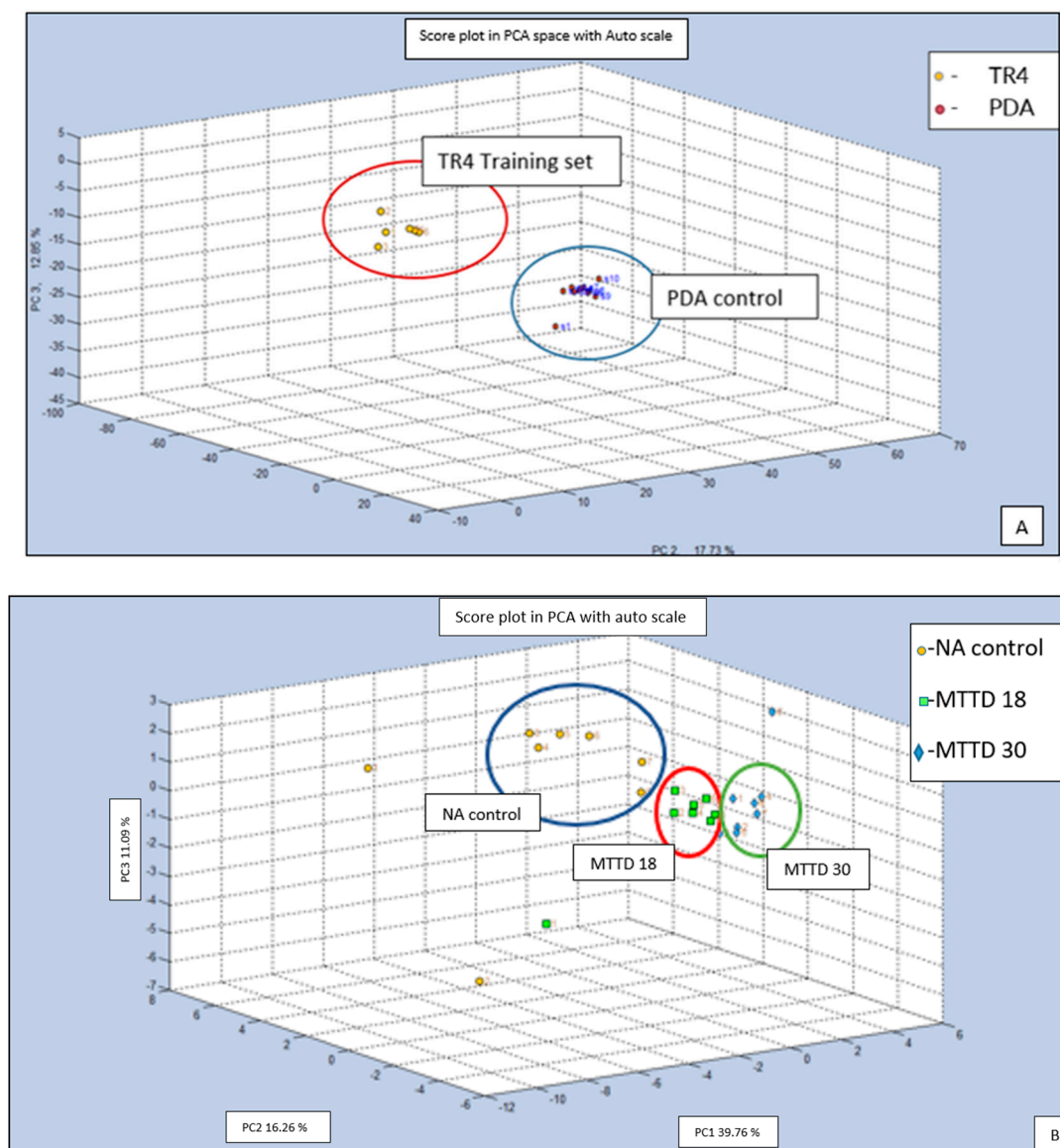


Figure 5. MSEM 160 PCA plot for (A) TR4 cultures and empty PDA control, (B) MTTD 18, MTTD 30 cultures and NA control.

3.2.2. GC-MS Analysis

Total ion chromatogram of the VOCs collected using SPMEs from an empty bag, media, bacterial cultures, infected and healthy banana pseudo-stems and mango leaves are presented in Figures 6 and 7. Although all VOCs from all samples were collected over two trials, only the peaks of the healthy and infected samples are presented in the Figures. A summary of all the VOCs detected is given in Table 5.

Compounds 1 to 4 (Table 5; blue squares) are volatiles collected from empty plastic bags and were one of the most consistently recovered VOC across all samples. Interestingly, none of the VOCs from the media (PDA) and living cultures (Tc or Ms) appear in the infected plant samples. This indicates that the VOCs released by infected plants differ greatly from those of the bacteria that cause the disease.

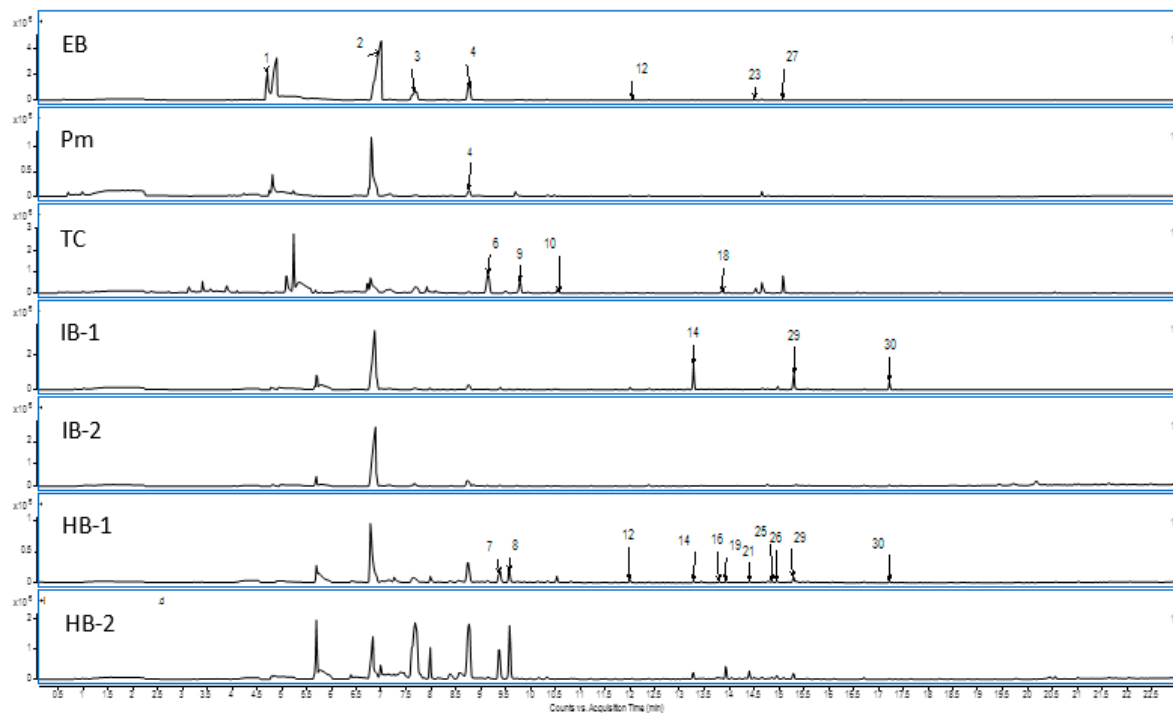


Figure 6. Chromatogram of samples related to Panama disease (TR4) in banana pseudo-stems EB: Empty bag, Pm: PDA media, Tc: TR4 culture, IB: Infected banana pseudo-stems and HB: Health banana pseudo-stems.

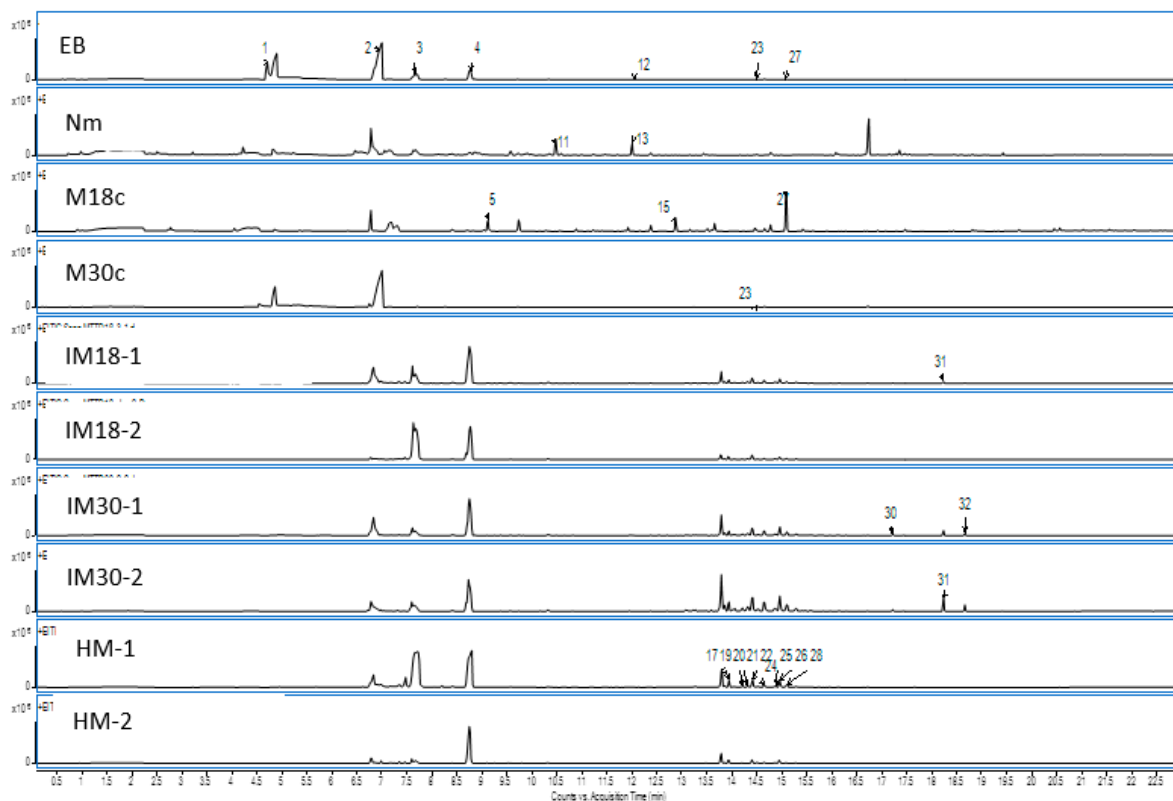


Figure 7. Chromatogram of samples related to Mango Twig Tip Dieback disease (MTTD) in mango leaves. EB: Empty bag, Nm: NA media, M18c: MTTD18 culture, M30c: MTTD30 culture, HM: Health mango leaves, IM18: Infected mango leaves with MTTD18, IM30: Infected mango leaves with MTTD30.

Table 5. Summary of VOCs identified from samples by using SPME-GC-MS.

RT(min).	No.	ID	EB	Pm	Tc	HB	IB	Nm	M18c	M30c	HM	IM18	IM30
4.7	1	Dimethylacetamide	+										
6.9	2	Phenol	+	+	+	+		+	+	+		+	
7.7	3	2-Ethyl-p-xylene	+		+	+	+			+	+	+	
8.8	4	p-(1-Propenyl)-toluene	+							+	+	+	
9.1	5	β-Phenylethanol							+				
9.2	6	p-Ethylanisole			+								
9.4	7	Cosmene				+							
9.6	8	Neo-allo-ocimene				+							
9.8	9	p-Vinylanisole			+								
10.5	10	6-Methyl-3,5-heptadiene-2-one			+								
10.5	11	Dodecane						+					
12.0	12	Indole	+			+							
12.0	13	Tridecane						+					
12.3	14	Copaene				+	++						
12.9	15	2,4-Toluene diisocyanate							+				
13.8	16	Chrysanthenone				+	+						
13.8	17	α-Gurjunene									+	+	+
13.9	18	α-Cedrene			+								
13.9	19	Caryophyllene				+	+				+	+	+
14.2	20	γ-Gurjunene									+	+	+
14.3	21	Humulene				+	+				+	+	+
14.3	22	Isoledene									+	+	+
14.5	23	2,6-Di-tert-butylbenzoquinone	+							+			
14.7	24	Bulnesene									+	+	+
14.9	25	Aromadendrene				+	+				+	+	+
15.0	26	Viridiflorene				+	+				+	+	+
15.1	27	BHT	+	+					+				
15.1	28	Guaiene									+	+	+
15.3	29	Calamenene				+	++						
17.2	30	Cadalene				+	+				+	++	++
18.2	31	Guaiazulene									+	++	++
18.7	32	3-Isopropyl-1,1'-biphenyl									+	++	++

EB: Empty bag, Pm: PDA media, Tc: TR4 culture, HB: Health banana pseudo-stems, IB: Infected banana pseudo-stems, Nm: NA media, M18c: MTTD18 culture, M30c: MTTD30 culture, HM: Healthy mango leaves, IM18: Infected mango leaves with MTTD18, IM30: Infected mango leaves with MTTD30. + indicates detection of the compound, and ++ indicates detection of high concentration.

Volatile 6 and 18 were unique to the pure culture of TR4 (Table 5; red squares). Ethylanisole and α-Cedrene were 2 of 6 VOCs produced by seven *F. oxysporum* strains [29]. The VOCs in infected and healthy banana pseudo-stems were similar (Table 5; green squares). However, compounds 14 and 29 (Copaene and Calamenene) were released in higher concentrations by infected samples compared to healthy samples.

Healthy and infected mango leaves also exhibit similar behaviour, with the VOCs released by healthy and infected mango leaves being similar, and the concentrations of compounds 30–32 were detected with high concentrations in the infected mango leaves (Table 5; brown squares). A group of sesquiterpenes was found to be common between healthy and infected banana pseudo-stems and mango leaves; for example, compounds 19, 21, 25 and 26 in Table 5 are present in all four samples (Table 5; yellow squares).

3.3. Differentiating Healthy and Infected Substrates

3.3.1. Electronic Noses

C 320 was only 54% accurate in identifying healthy and MTTD-infected leaves and 75% accuracy in differentiating healthy and TR4-positive pseudo stems (Tables 6 and 7). None of the classes had obtained the correct total sample identification. This was out of six replicates for each class. The overall result suggests the suitability of the C 320 device for qualitative measurements rather than quantitative analysis.

Table 6. Cross-validation results of Healthy, MTTD 18 and MTTD 30-infected mango leaves using C320.

Trained as	Healthy	MTTD 18	MTTD 30	% Accuracy
No. of correct identification/6 (total no. of replicates)	4	4	5	54.17

Table 7. Cross-validation results of Healthy banana pseudo-stem and TR4 infected pseudo-stem samples.

Trained as	Healthy	TR4 Positive	% Accuracy
No. of correct identification/6 (total no. of replicates)	4	5	75

C 320 was capable of differentiating healthy mango leaves from randomly collected MTTD-infected leaves. Figure 8A demonstrates how these samples are separated in a PCA plot. All healthy samples congregated into a cluster, while MTTD-infected samples were scattered in the plot. Figure 8B shows less capability of C 320 in discriminating healthy banana pseudo stems from TR4 infected samples.

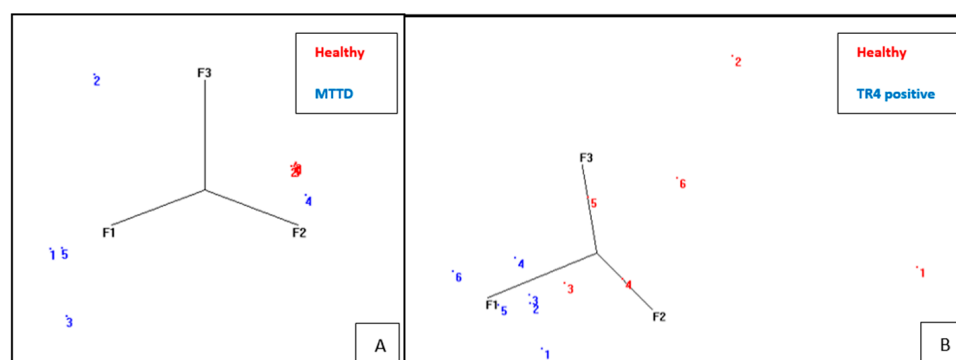


Figure 8. PCA plot for (A) Healthy and MTTD infected mango leaves, (B) Healthy and TR4 positive banana pseudo stem samples in C 320.

The PEN 3 successfully differentiated three classes of MTTD 18 infected mango leaves, MTTD 30 infected mango leaves, and healthy/control leaves with a variance of 86.06%, which is an excellent value of separation. Figure 9A shows a variance of 71.59% in the first main axis and 14.47% in the second main axis within the three classes. The distance between the control and MTTD-infected samples was higher than the distance between two MTTD-infected varieties. This result suggests that infected samples have similar volatile characteristics, which are different from the volatiles of the control. The distinct separation of two classes, healthy and TR4 infected pseudo-stems, was observed in the PCA plot with a high variance of 84.96%, as shown in Figure 9B. The result demonstrates the high efficacy of the PEN 3 device in differentiating TR4 and MTTD infections from healthy samples and, hence, the suitability of the device in pathogen detection.

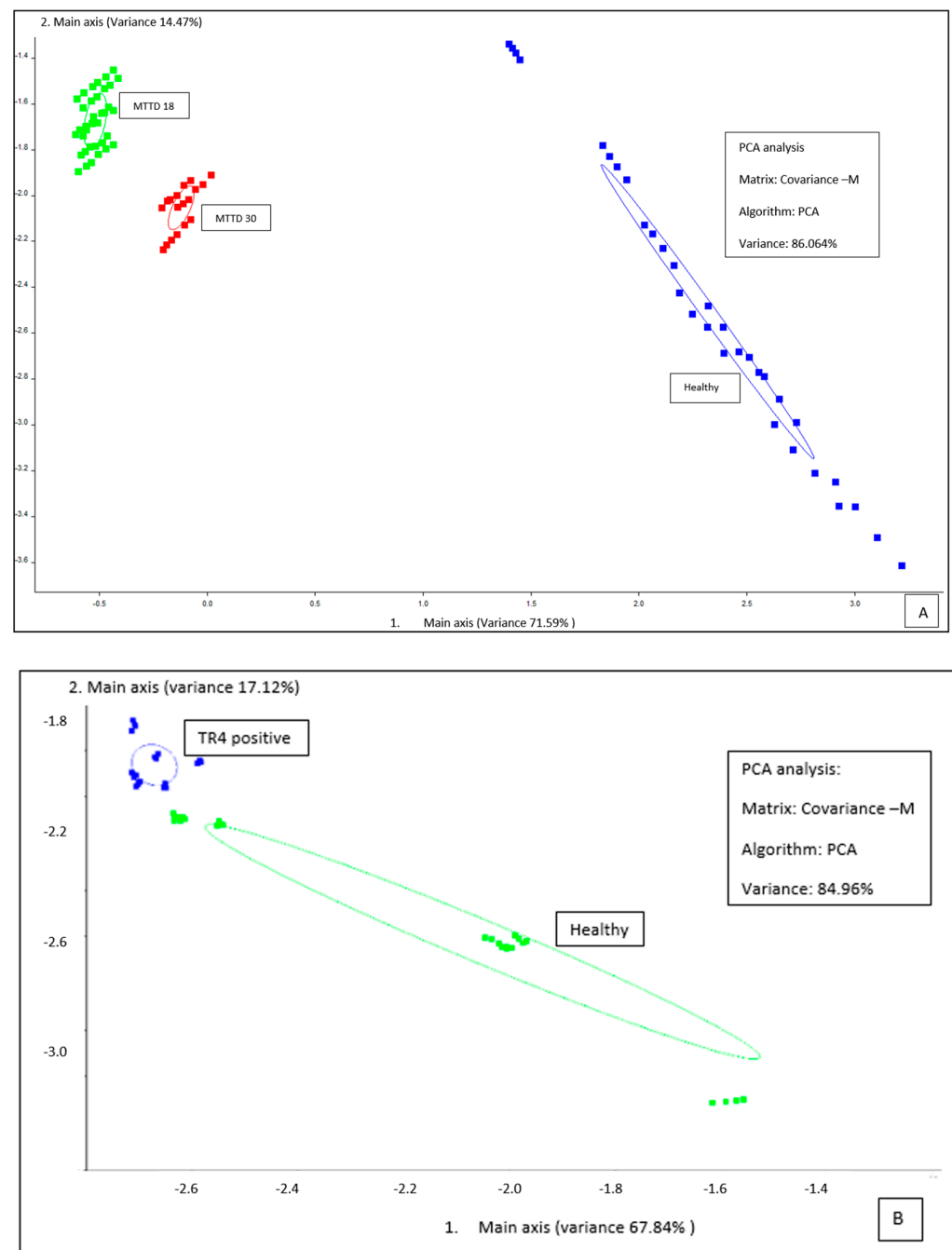


Figure 9. PCA plot for (A) MTTD infected samples and healthy mango leaves, (B) TR4 positive banana pseudo-stem samples and healthy pseudo-stem samples in PEN 3.

The distinct separation of three classes of leaf samples (control, MTTD 18 and MTTD 30) was observed in the PCA score plot by CD analysis, as shown in Figure 10A. The results suggest that MSEM 160 can identify and differentiate volatiles of healthy mango leaves from infected samples. Furthermore, the result provides adequate evidence for the high efficacy of the device in differentiating two infected MTTD varieties. Figure 10B shows significant discrimination of two classes, TR4 infected pseudo stems and healthy pseudo stem samples. PCA plot of MSEM 160 thoroughly demonstrates the capability of the device in advanced pathogen identification, hence suitable for further development of the technology to be used in the field.

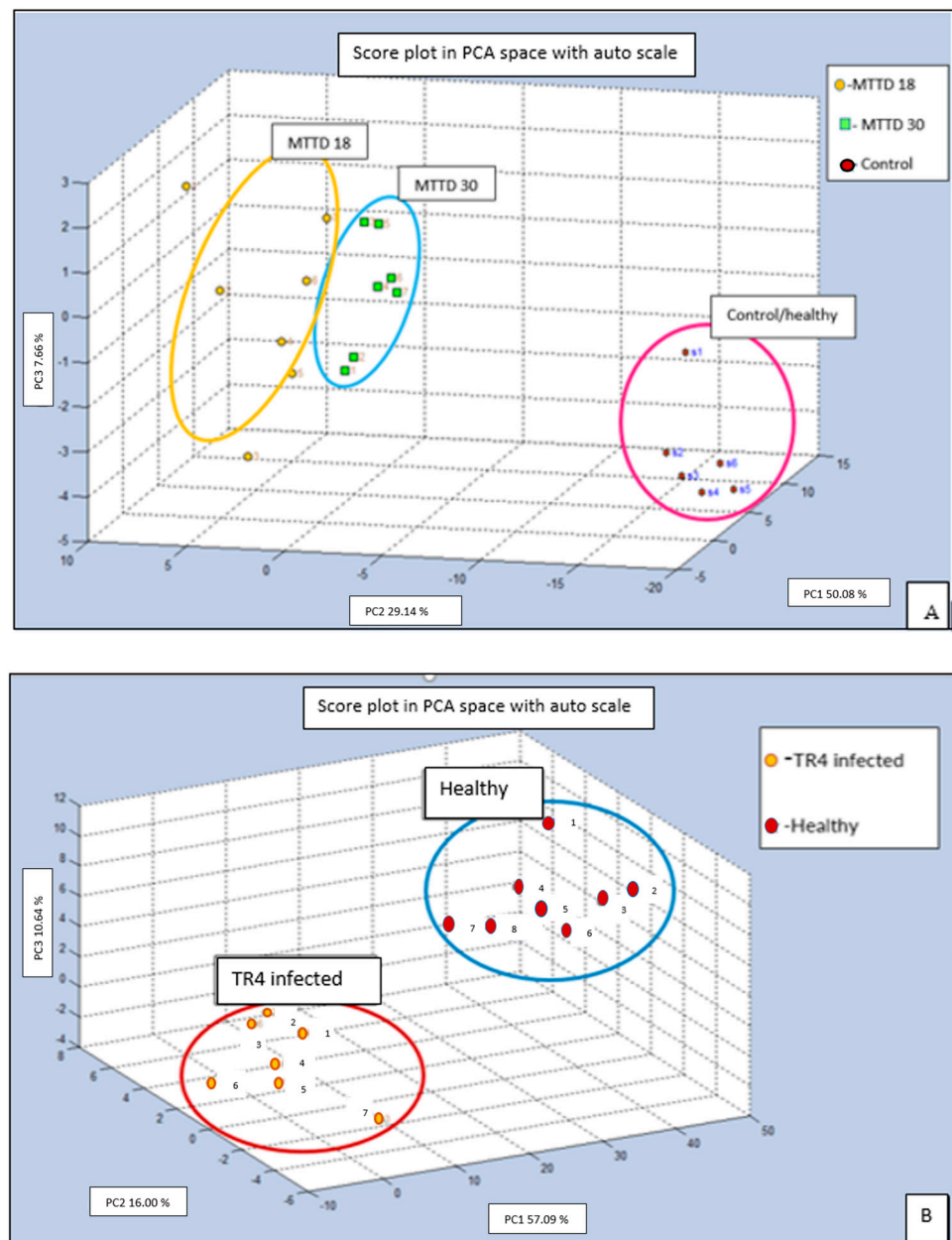


Figure 10. PCA plot for MSEM 160 (A) MTTD 18, MTTD 30 and control/healthy mango leaves, (B) TR4 infected and healthy banana pseudo-stems.

3.3.2. GC-MS Analysis

The PCA resemblance plot of the VOCs identified by GC-MS analysis is shown in Figure 11. The VOCs were collected from two types of media and the M18c bacterial culture, segregated from the Tc- and M30c-cultures. For the infected plant substrates, GC-MS analysis clustered the health and infected plant samples of both mango and banana pseudo-stem together.

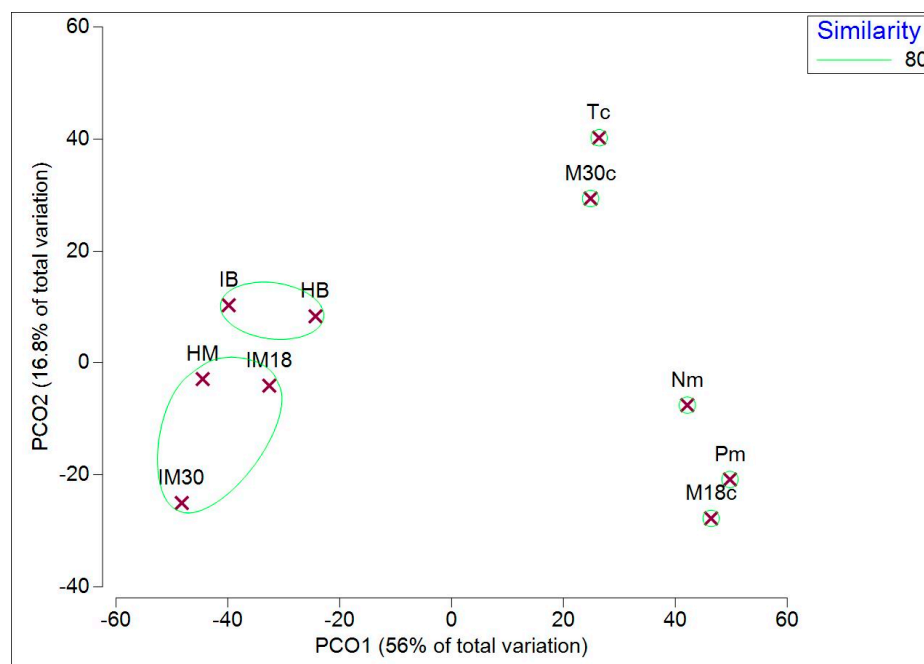


Figure 11. PCA score plot of GC-MS VOCs identified in all the samples. Pm: PDA media, Tc: TR4 culture, HB: Health banana pseudo-stems, IB: Infected banana pseudo-stems, Nm: NA media, M18c: MTTD18 culture, M30c: MTTD30 culture, HM: Health mango leaves, IM18: Infected mango leaves with MTTD18, IM30: Infected mango leaves with MTTD30.

4. Conclusions

This is a pilot study of checking E-noses for disease identification in controlled laboratory conditions. Results from this study evaluate the capacity of three E-noses to discriminate between healthy and infected plants. C 320 was 54% accurate in identifying healthy and MTTD-infected leaves, and 75% accurate in differentiating healthy and TR4-positive pseudo stems. PEN 3 presented an 86.06% variance between two MTTD-infected and healthy leaves. Also, a variance of 84.96% between TR4 positive and healthy pseudo stems was observed. MSEM 160, as the advanced device, has demonstrated clear separation of individual samples into particular classes where they belong (TR4, PDA, MTTD, healthy, etc.). Of the three devices, the C 320 would be considered unsuitable for quantitative studies. However, it is a hand-held, portable unit that can be used in field applications. The PEN 3 and MSEM 160 have quantitative capabilities to differentiate between synthetic and natural volatiles associated with pure plant pathogens and infected plant material. The limitation of these units is that they are not portable, are suited to a laboratory set-up, and require a separate computer to process and visualise data by a trained specialist. GC-MS analysis confirmed unique VOC fingerprints of pure pathogen cultures and infected plant materials such as Ethylanisole and α -Cedrene, but the VOCs from infected and healthy plant material were found to be the same, although varied in concentration. GC-MS result indicates that the VOCs released by infected plants are very different from those of the bacteria that cause the disease.

Our results demonstrate that E-nose technology was more efficient and rapid than GC-MS in differentiating between healthy and diseased plant substrates. Although E-noses have a few limitations, such as sensitivity to undesired environmental odours and humidity, these devices appear to be more suitable for detecting pathogen and plant disease volatiles. Therefore, this technology can potentially increase the efficiency of pathogen and plant disease diagnosis in controlled conditions and would support rapid identification.

Supplementary Materials: The following supporting information can be downloaded at <https://www.mdpi.com/article/10.3390/chemosensors12070117/s1>: Figure S1. Linear Discriminant Analysis plot for individual classes of concentration and total concentration series data obtained from PEN 3 device. (A) 3-Methyl-2-butanol, (B) Isoamyl isovalerate. Figure S2. Linear Discriminant Analysis Plots of data obtained from the PEN 3 device. (A) 3M2B total series vs IAIV total series, (B) Mixture of 1 ppm 3M2B and IAIV against pure 3M2B and IAIV of 1 ppm. Figure S3. CD analysis score plots in PCA of data obtained from MSEM 160 device. (A) 3-Methyl-2-butanol total concentration series, (B) Mixture of 1 ppm 3M2B and IAIV.

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References

1. Sankaran, S.; Mishra, A.; Ehsani, R.; Davis, C. A Review of Advanced Techniques for Detecting Plant Diseases. *Comput. Electron. Agric.* **2010**, *72*, 1–13. [\[CrossRef\]](#)
2. Zee, F.; Judy, J.W. Micromachined Polymer-Based Chemical Gas Sensor Array. *Sens. Actuators B Chem.* **2001**, *72*, 120–128. [\[CrossRef\]](#)
3. Cui, S.; Ling, P.; Zhu, H.; Keener, H. Plant Pest Detection Using an Artificial Nose System: A Review. *Sensors* **2018**, *18*, 378. [\[CrossRef\]](#)
4. Webster, C.G.; Wylie, S.J.; Jones, G.K. Diagnosis of plant viral pathogens. *Curr. Sci.* **2004**, *86*, 1604–1607.
5. Kwon, J.Y.; Ryu, K.H.; Choi, S.H. Reverse Transcription Polymerase Chain Reaction-Based System for Simultaneous Detection of Multiple Lily-Infecting Viruses. *Plant Pathol. J.* **2013**, *29*, 338–343. [\[CrossRef\]](#)
6. Sanjay, M.; Kalpana, B. Early Mass Diagnosis of Fusarium Wilt in Banana Cultivations Using an E-Nose Integrated Autonomous Rover System. *Int. J. Appl. Sci. Biotechnol.* **2017**, *5*, 261–266. [\[CrossRef\]](#)
7. Raza, W.; Wang, J.; Wu, Y.; Ling, N.; Wei, Z.; Huang, Q.; Shen, Q. Effects of Volatile Organic Compounds Produced by *Bacillus Amyloliquefaciens* on the Growth and Virulence Traits of Tomato Bacterial Wilt Pathogen *Ralstonia Solanacearum*. *Appl. Microbiol. Biotechnol.* **2016**, *100*, 7639–7650. [\[CrossRef\]](#)
8. Baldwin, I.T.; Halitschke, R.; Paschold, A.; Von Dahl, C.C.; Preston, C.A. Volatile Signaling in Plant-Plant Interactions: “Talking Trees” in the Genomics Era. *Science* **2006**, *311*, 812–815. [\[CrossRef\]](#)
9. Zou, H.-Q.; Li, S.; Huang, Y.-H.; Liu, Y.; Bauer, R.; Peng, L.; Tao, O.; Yan, S.-R.; Yan, Y.-H. Rapid Identification of *Asteraceae* Plants with Improved RBF-ANN Classification Models Based on MOS Sensor E-Nose. *Evid.-Based Complement. Altern. Med.* **2014**, 1–6. [\[CrossRef\]](#)
10. Paré, P.W.; Tumlinson, J.H. Plant Volatiles as a Defense against Insect Herbivores. *Plant Physiol.* **1999**, *121*, 325–332. [\[CrossRef\]](#)
11. Maffei, M.E. Sites of Synthesis, Biochemistry and Functional Role of Plant Volatiles. *S. Afr. J. Bot.* **2010**, *76*, 612–631. [\[CrossRef\]](#)
12. Cui, S.; Inocente, E.A.A.; Acosta, N.; Keener, H.M.; Zhu, H.; Ling, P.P. Development of Fast E-Nose System for Early-Stage Diagnosis of Aphid-Stressed Tomato Plants. *Sensors* **2019**, *19*, 3480. [\[CrossRef\]](#)
13. Gherghel, S.; Morgan, R.M.; Arrebola-Liébanas, J.; Romero-González, R.; Blackman, C.S.; Garrido-Frenich, A.; Parkin, I.P. Development of a HS-SPME/GC-MS Method for the Analysis of Volatile Organic Compounds from Fabrics for Forensic Reconstruction Applications. *Forensic Sci. Int.* **2018**, *290*, 207–218. [\[CrossRef\]](#)
14. Ouyang, G.; Pawliszyn, J.A. Critical Review in Calibration Methods for Solid-Phase Microextraction. *Anal. Chim. Acta* **2008**, *627*, 184–197. [\[CrossRef\]](#)
15. Wu, Z.; Wang, H.; Wang, X.; Zheng, H.; Chen, Z.; Meng, C. Development of Electronic Nose for Qualitative and Quantitative Monitoring of Volatile Flammable Liquids. *Sensors* **2020**, *20*, 1817. [\[CrossRef\]](#)

16. Murray, J.; Delahunty, C.; Baxter, I. Descriptive Sensory Analysis: Past, Present and Future. *Food Res. Int.* **2001**, *34*, 461–471. [[CrossRef](#)]
17. Xu, X.; Tian, F.; Yang, S.; Li, Q.; Yan, J.; Machacek, J. A Solid Trap and Thermal Desorption System with Application to a Medical Electronic Nose. *Sensors* **2008**, *8*, 6885–6898. [[CrossRef](#)]
18. Alphus Dan Wilson, A.D. Recent Progress in the Design and Clinical Development of Electronic-Nose Technologies. *NDD* **2016**, *15*, 15–27. [[CrossRef](#)]
19. Lee, S.J.; Noble, A.C. Characterization of Odor-Active Compounds in Californian Chardonnay Wines Using GC-Olfactometry and GC-Mass Spectrometry. *J. Agric. Food Chem.* **2003**, *51*, 8036–8044. [[CrossRef](#)]
20. Vandenvelde, S.; Nevens, F.; Vanhee, P.; Vansteenbergh, D.; Quirynen, M. GC-MS Analysis of Breath Odor Compounds in Liver Patients. *J. Chromatogr. B* **2008**, *875*, 344–348. [[CrossRef](#)]
21. Mota, I.; Teixeira-Santos, R.; Cavaleiro Rufo, J. Detection and Identification of Fungal Species by Electronic Nose Technology: A Systematic Review. *Fungal Biol. Rev.* **2021**, *37*, 59–70. [[CrossRef](#)]
22. Cellini, A.; Biondi, E.; Blasioli, S.; Rocchi, L.; Farneti, B.; Braschi, I.; Savioli, S.; Rodriguez-Estrada, M.T.; Biasioli, F.; Spinelli, F. Early Detection of Bacterial Diseases in Apple Plants by Analysis of Volatile Organic Compounds Profiles and Use of Electronic Nose: Early Detection of Bacterial Diseases by Volatile Organic Compounds Profiling. *Ann. Appl. Biol.* **2016**, *168*, 409–420. [[CrossRef](#)]
23. Bonah, E.; Huang, X.; Aheto, J.H.; Osae, R. Application of Electronic Nose as a Non-Invasive Technique for Odor Fingerprinting and Detection of Bacterial Foodborne Pathogens: A Review. *J. Food Sci. Technol.* **2020**, *57*, 1977–1990. [[CrossRef](#)]
24. Loulier, J.; Lefort, F.; Stocki, M.; Asztemborska, M.; Szmigielski, R.; Siwek, K.; Grzywacz, T.; Hsiang, T.; Ślusarski, S.; Oszako, T. Detection of Fungi and Oomycetes by Volatiles Using E-Nose and SPME-GC/MS Platforms. *Molecules* **2020**, *25*, 5749. [[CrossRef](#)]
25. Daly, A.; Walduck, G. Fusarium Wilt of Bananas (Panama Disease)(*Fusarium oxysporum* f. sp. cubense). *Agnote* **2006**, *151*, 1–5.
26. Jolliffe, I. *Principal Component Analysis*, 2nd ed.; Springer: New York, NY, USA, 2002; pp. 111–149.
27. Sun, Y.; Zheng, Y. Prediction of tomato plants infected by fungal pathogens at different disease severities using E-nose and GC-MS. *J. Plant Dis. Prot.* **2024**, *131*, 835–846. [[CrossRef](#)]
28. Refaeilzadeh, P.; Tang, L.; Liu, H. Cross-Validation. In *Encyclopedia of Database Systems*; Liu, L., Özsu, M.T., Eds.; Springer: Boston, MA, USA, 2009; pp. 532–538. [[CrossRef](#)]
29. Li, N.; Alfiky, A.; Wang, W.; Islam, M.; Nourollahi, K.; Liu, X.; Kang, S. Volatile compound-mediated recognition and inhibition between *Trichoderma* biocontrol agents and *Fusarium oxysporum*. *Front. Microbiol.* **2018**, *9*, 2614. [[CrossRef](#)]

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